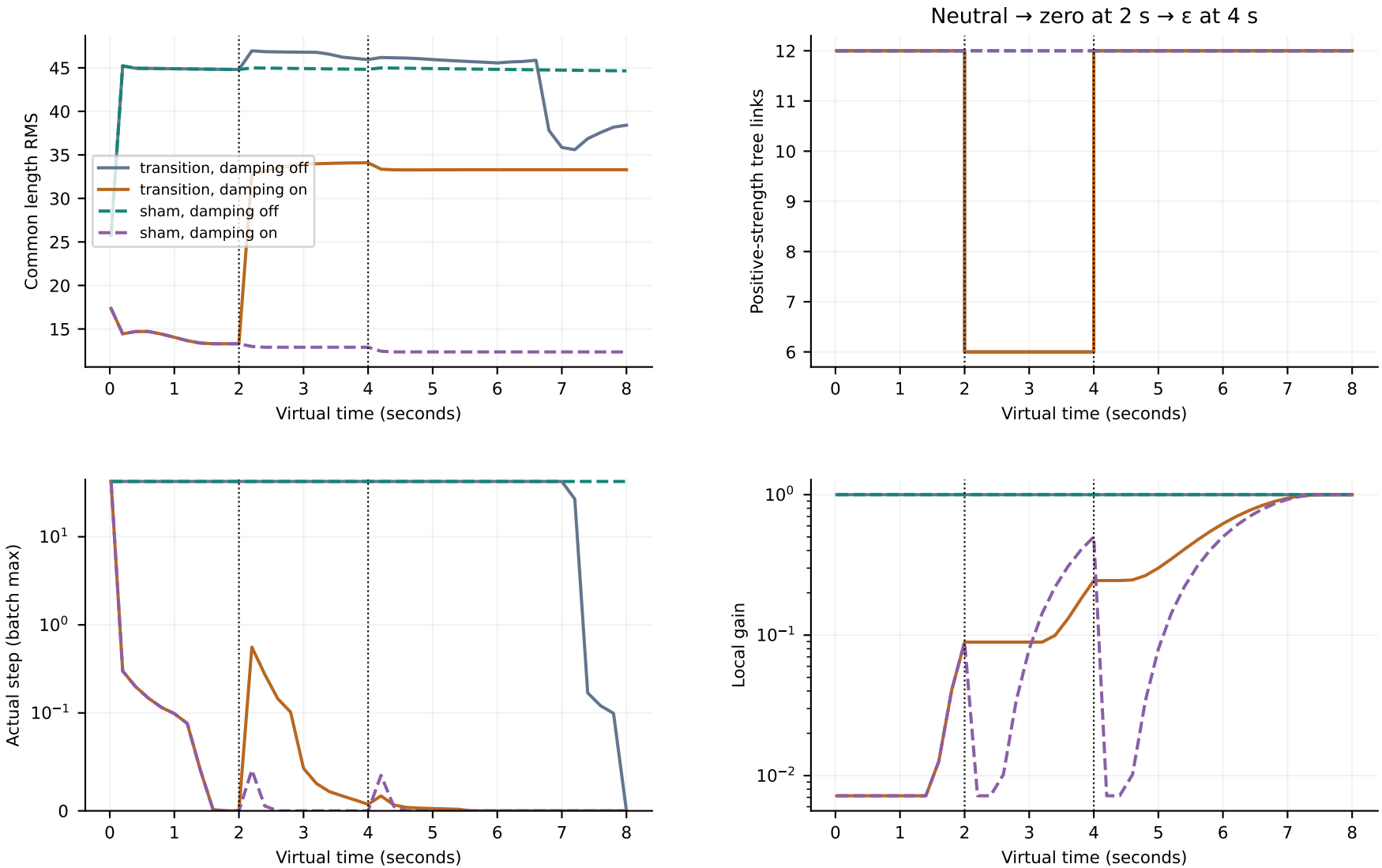
F4. Zero transitions and a matched wake-only control



F4. Seed 0xdecafbad, movable leaf-linked trees. Weights change only after batches 100 and 200; both events wake global cooling. The sham keeps neutral weights with identical wakes, isolating weight changes from reheating. Positive ε restores branch participation even when its force is tiny. The membership plot includes exact change events; movement/gain are sampled batches. All integration steps contribute to totals.